Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII SEMESTER B.TECH END SEMESTER EXAMINATION, NOVEMBER 2024

Medical Imaging [BME 4058]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Instructions to Candidates:
[i] Answer ALL questions
[ii] Missing data may be suitably assumed
[iii] No Formula book will be provided

1A)	Explain the steps involved in gamma camera image formation. How do the collimator and photomultiplier tubes contribute to the overall process?	(3)
1B)	Examine how varying the number of projections in the Filtered Backprojection (FBP) algorithm affects reconstruction accuracy. What artifacts might arise, and how can they be minimized?	(3)
1C)	Evaluate the impact of different filter choices, such as Ramp, Shepp-Logan, and Cosine, on the quality of CT image reconstructions. How does each filter influence image sharpness and noise levels?	(4)
2A)	Analyze the differences in reconstruction outcomes when using the Algebraic Reconstruction Technique (ART) versus Simultaneous Iterative Reconstruction Technique (SIRT) for CT imaging. How do these methods impact noise levels and image stability?	(5)
2B)	Evaluate the impact of altering TR and TE on the contrast in T1 and T2 weighted images. In what clinical scenarios might you prioritize T1 weighted imaging over T2 weighted imaging?	(3)
2C)	Analyse how the strength of the main magnetic field (B_0) influences the precessional frequency of hydrogen protons. How would a higher BO field strength affect signal-to-noise ratio and spatial resolution?.	(2)

3A)	Examine the factors that influence slice thickness in MRI, specifically the role of RF pulse bandwidth and gradient strength. How does adjusting these parameters affect both the thickness and resolution of the slice?	(3)
3B)	Interpret how the frequency encoding gradient affects both image acquisition speed and image detail. If an imaging protocol requires a balance between resolution and speed, how might adjustments to the frequency encoding gradient be utilized?	(3)
3C)	Evaluate the influence of increasing the phase encoding gradient steps on image quality and scan duration. In what scenarios would maximizing phase encoding steps be essential?	(4)
4A)	Explain why a rephasing gradient is applied after slice selection in MRI. How does rephasing correct the effects of dephasing, and why is this step important for achieving a uniform signal from the selected slice?	(4)
4B)	Construct a sequential overview of the image reconstruction process from k-space, detailing how data acquisition, inverse Fourier transform, and normalization contribute to final image clarity and diagnostic utility. Why is each of these steps critical for producing a usable MRI image?	(3)
4C)	Interpret the significance of the central region in a k-space image that appears with high-amplitude signals compared to the peripheral regions. What kind of information does the central region primarily provide, and how would this impact the overall contrast of the MRI image?	(3)
5A)	Illustrate the process by which acoustic impedance differences between tissues create reflections in ultrasound imaging.	(4)
5B)	Assess the principles of pulse-echo ultrasound imaging and determine the effects of pulse repetition frequency (PRF) on imaging depth and temporal resolution.	(3)
5C)	With the help of an algorithm explain the sequence of steps involved in ultrasound image reconstruction from signal acquisition to final display.	(3)

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